

The effects of mother-father relationships on children's social-emotional competence: The chain mediating model of parental emotional expression and parent-child attachment

Yu Tang^{a,*}, Shaomei Li^{a,*}, Lin Ma^a, Yuxin Zheng^a, Yan Li^b

^a Faculty of Education, Shaanxi Normal University, Xi'an, Shaanxi Province, China

^b Zibo Normal College, Zibo, Shandong Province, China

ARTICLE INFO

Keywords:

Mother-father relationships
Social-emotional competence
Parental emotional expression
Parent-child attachment

ABSTRACT

Mother-father relationships, as one of the key elements of the marriage subsystem, play an essential function in the improvement of social-emotional competence. The study used social learning theory as an analytic framework to consider the potential mechanisms by which mother-father relationships influence social-emotional competence. Therefore, 802 children and their parents from 54 classes in 27 preschools in China were selected to research the chain-mediating function of parental emotional expression and parent-child attachment in mother-father relationships affecting social-emotional competence. The results indicated that (1) Mother-father relationships significantly and positively predicted social-emotional competence. (2) Parental emotional expression and parent-child attachment separately fully mediated the relationship between mother-father relationships and social-emotional competence. (3) Parental emotional expression and parent-child attachment were chain mediators between mother-father relationships and social-emotional competence. This study highlights the important contribution of mother-father relationships in the improvement of social-emotional competence. Hence, parents are supposed to realize the significance of mother-father relationships in the cultivation of social-emotional competence. Meanwhile, parents should express emotions rationally in the family. They should give their children effective support and companionship to form a high-quality parent-child attachment, which ultimately affects the improvement of social-emotional competence.

1. Introduction

Social-emotional competence plays a vital position in individual development and has gotten more notice in recent years. The improvement of social-emotional competence can be well known by considering family relationships. Instead, family relationships arise from the interaction of different subsystems in the family. As one of the key elements of the marriage subsystem, mother-father relationships are central to family relationships. It plays a crucial part in the acquisition of social-emotional competence. Nevertheless, researchers have few agreed on answering or explaining how mother-father relationships impact social-emotional competence. According to social learning theory, variables such as parental emotional expression and parent-child attachment may be potential variables connecting mother-father relationships to social-emotional competence. The mechanism of influence of mother-father relationships on social-emotional competence is likely to be a

mediating chain. Therefore, this study explored the potential mechanism by which mother-father relationships as a dynamic factor influence social-emotional competence and the chain mediating function of parental emotional expression and parent-child attachment. This provides new ideas to promote the improvement of social-emotional competence.

1.1. Theoretical analytical framework

The essence of children's learning is social observation and imitation. Focusing on and exploring social-emotional competence requires a systematic analysis that places children in the context of their social learning and lives. Considering the logical relationships between different factors can provide a doctrinal basis for describing, interpreting, and predicting the relationships between variables. Social learning theory offers the possibility of exploring the mechanisms by which

* Corresponding authors at: Education College, Shaanxi Normal University, No.199, Chang'an South Road, Yanta, Xi'an, China.

E-mail addresses: tangyu@snnu.edu.cn (Y. Tang), shaomei@snnu.edu.cn (S. Li).

<https://doi.org/10.1016/j.childyouth.2023.107227>

Received 7 June 2023; Received in revised form 11 October 2023; Accepted 12 October 2023

Available online 13 October 2023

0190-7409/© 2023 Elsevier Ltd. All rights reserved.

children's abilities work concerning their social learning. As a projection of social life, family life is the primary arena for children's participation in social life. Parental factors, as one of the most active variables in the family, are key variables in children's social-emotional competence. From this theoretical perspective, research should focus on the impact of parental factors in family life on children's possible developmental potential, especially on improving social-emotional competence.

Social learning theory proposes that observers will extract the common attributes represented from different exemplary responses through imitation. This ultimately results in rules that produce behaviors with similar structural features (Bandura and Walters, 1977). At the same time, higher-status archetypes are more effective than lower-status archetypes in motivating individuals to produce similar behaviors. For children, parents are more powerful, capable, and empowered. This makes it easier for parents to become prototypes for children to imitate. As children attempt to improve their social-emotional competence, parental social-emotional competence becomes the primary object of imitation for children. In the process, children form a basic mental representation of social-emotional competence through conscious or unconscious observation of their parental social-emotional competence. In turn, children will develop their social-emotional competence based on these cognitive experiences. That is, in a causal relationship, we can hold parental social-emotional competence as an antecedent and children's social-emotional competence as the final consequence. Mother-father relationship quality, parental emotional expression quality, and parent-child attachment are all manifestations of parental social-emotional competence. From a social learning perspective, there is also a more complex relationship between the intrinsic factors of parental social-emotional competence. Specifically, first, environmental and behavioral factors are interdependent and interact with each other. However, mother-father relationship quality, as an important "environmental" factor, continuously influences the "behavioral" factors of parents. Specifically, in the process of creating the relationship, both parents will display attitudes or value judgments about behavior, resulting in different qualities of emotionally expressive behavior. That is, "environmental" factors contribute to "behavioral" factors. Put differently, we can hold mother-father relationship quality as an antecedent and parental emotional expression quality as the final consequence. Second, social learning theory focuses on the results of interactions between behavioral and human factors. It suggests that factors interact with each other with varying degrees of influence and that patterns of interaction are in a constant state of flux. In the process of parents expressing their emotions, parents may intentionally or unintentionally include children in their interactions, resulting in different "behavioral" factors. These factors flow into the parent-child interaction as derivatives of the parental emotional expression. When children perceive these factors, they respond accordingly. In turn, the parent acquires a new "human" factor, that is, parent-child attachment. That is, in the causal sequence, we can hold parental emotional expression quality as an antecedent and parent-child attachment as the final consequence. Third, whether or not an environmental factor affects an individual depends on whether or not the individual acts to activate it. This is the main point of social learning theory. That is, it is only when children actively and proactively interact with mother-father relationships as environmental factors that they are likely to be influenced by their qualities and, consequently, to behave differently. In other words, children will resonate with mother-father relationships by actively resonating with them, thereby accepting their influence, absorbing their potential, and creating new generations. In turn, the child will acquire a new type of social relationship, that is, parent-child attachment. Thus, in causation, we can hold mother-father relationship quality as an antecedent and parent-child attachment as the final consequence. This similarly implies that mother-father relationship quality can influence parent-child attachment not only directly, but also indirectly through parental emotional expression quality.

Not only is there a complex relationship between the underlying

factors of parental social-emotional competence, they can each have a direct impact on children's social-emotional competence. Specifically, first, the activation and persistence of behavior needs to be understood through the ongoing interactions between people and situational factors. Social learning theory suggests that the environment can exert a powerful hold on the developmental formation of an individual's behavior. Mother-father relationship quality, an important factor in the family, also influences children's social-emotional competence. On the one hand, mother-father relationship quality can serve as an external condition that influences children's social-emotional competence and creates an appropriate learning environment for their development. On the other hand, mother-father relationship quality can also serve as a source of information. Mother-father relationship quality determines the extent to which parents convey to children a diversity of ideas and behavioral attitudes about social activities. This diversity of perceptions and behaviors allows children to acquire their parents' social attitudes and values and to develop family-specific social-emotional competence. Therefore, we can hold mother-father relationship quality as an antecedent and children's social-emotional competence as the final consequence. Second, alternative reinforcement is an essential point of view in social learning theory. It implies that the individual does not need to be directly involved in the learning activity. Children can acquire similar expressions simply by observing the behavior of others and its consequences (Brady et al., 2021). Emotionally expressive behaviors that occur during parental interactions provide children with indirect emotional experiences. This enables children to establish emotional responses to relevant matters in their environment. Specifically, parental emotional expression quality influences the outcome of children's engagement in alternative reinforcement. That is, by observing high-quality parental emotional expression, children find the behavior praised and enjoyable. This reinforces the likelihood that children will exhibit high-quality emotionally expressive behaviors. In the process, children acquire patterns of emotional expression present in the family through alternative reinforcement. In turn, children internalize this information as a way of coordinating their interactions with the outside world and continue to learn to express emotions rationally and construct social-emotional competence. In other words, from the perspective of the causal sequence, we can hold parental emotional expression quality as an antecedent and children's social-emotional competence as the final consequence. Third, all behaviors of individuals are formed under the influence of the social environment and through observational learning of others' behaviors. In the process of observational learning, individuals not only need to notice the "role model" behavior but also need to "reproduce" the behavior (Salali et al., 2019). That is, the individual should transform the learned symbols into behaviors so that the results of learning are taken into account in their actions, behaviors, and perceptions. That is to say, in the process of parent-child attachment formation, children will learn the corresponding behavioral symbols by observing their parents' interaction patterns with them. In turn, children will transform the learned symbols into their behavior and complete the reproduction of the behavior. The behavior will be reinforced or punished to varying degrees in the process of forming parent-child attachment, thus regulating and correcting children's social tendencies and abilities. As a result, in causation, we can hold parent-child attachment as an antecedent and children's social-emotional competence as the final consequence.

In summary, social-emotional competence is constructed by individuals in the process of social learning. Social learning theory emphasizes the interaction of the individual, behavior, and environment, as well as the essential value of observational learning, alternative experiences, and so on. This can provide a doctrinal basis for describing, explaining, and predicting the relationships between mother-father relationship quality, emotional expression quality, parent-child attachment, and children's social-emotional competence. It also serves as a theoretical framework for analyzing the research hypotheses.

1.2. Social-emotional competence

Social-emotional competence is the acquisition and application by children of a core set of competencies related to an individual's social and emotional growth in the complicated situation of evolution and development (Li et al., 2023). Firstly, social-emotional competence could significantly predict a child's academic performance. That is, children with better social-emotional competence have a higher sense of belonging to school upon entry, better behaviors during school, and they have better performance in school. Conversely, children who present with social-emotional impairment are expected to have a higher likelihood of future problems such as low academic ability and achievement. Secondly, social-emotional competence is a crucial factor influencing an individual's psychological development. Improving social-emotional competence can reduce depression and anxiety. Also, social-emotional competence has a definite contribution to psychological factors such as psychological resilience, self-uniformity, decisiveness, and self-management. It assists children in identifying the state of themselves and others. Finally, social-emotional competence could support children in developing good interpersonal relationships. When a child has appropriate social-emotional skills, he or she tends to be able to transpose thinking, respect, and accept others' different views and opinions. Also, they will have better interaction skills, which include communicating and listening to others, and participating in society actively. Children are also able to analyze and solve problems better based on the consideration of self and other interests, demonstrate responsible decision-making behaviors, and build close emotional connections. In addition, social-emotional competence has a more prominent role in promoting disadvantaged children. It can compensate for the negative effects of disadvantage and play a compensatory role. In summary, social-emotional competence has a specific and decisive role in the development of the individual and the preparation of the child for a meaningful life.

Considering the improvement of social-emotional competence, current academic research on social-emotional competence broadly follows two paradigms. One is based on a pedagogical research perspective. The study takes the perspectives of constructivism theory and conceptual history to sort out the development of the concept and to explore the variations in social-emotional competence in different disciplines and different historical development periods. The other one uses the psychological research paradigm. Such research focuses on family variables such as parenting style, home environment, and other internal factors in the family on the improvement of social-emotional competence. Because they have different respective value orientations and theoretical presuppositions, there are problems such as isolation and fragmentation between the two types of perspectives and a lack of theoretical dialogue. This can be explained as follows. Firstly, the discursive-oriented research emphasizes theoretical constructs, which to a certain extent weakens the plasticity of social-emotional competence. While the empirical-based studies lack theoretical support and ignore the nature of social-emotional competence. Thus, the existing single research perspective tends to fall into binary isolation of either theory or empirical evidence. Secondly, social-emotional competence is formed through social observation and learning. It is influenced by the collective actions of individuals, the environment, and behavior. Current research only pays attention to the impact of diverse home elements on social-emotional competence but has not explored the effect of home factors on it from a social-learning perspective. Mother-father relationships are elements that exist in the home and have an irreplaceable effect on the improvement of social-emotional competence. Thus, we suggest focusing on the interaction between independent and dependent variables based on a social learning perspective, which has positive implications for the early cultivation of social-emotional competence.

1.3. Mother-father relationships and social-emotional competence

Mother-father relationships can be explained as the process of interaction and the degree of rapport between the individual in the father role and the individual in the mother role. Importantly, mother-father roles are constructed in social interactions with children. Thus, the essential characteristic that distinguishes mother-father relationships from other social relationships is that children are embedded in them. Mother-father relationships include two dimensions, marital conflict and marital support. The former is interested in the negative aspects of mother-father relationships, such as anxiety, violence, indifference, and breakdowns (Cundiff et al., 2015). The latter concerns the proactive elements of mother-father relationships, such as support, attachment, and satisfaction (Lavner et al., 2016).

According to social learning theory, mother-father relationships can act as an environmental stimulus. It conveys information about the likelihood of social actions or events, which in turn influences social-emotional competence. From a supportive standpoint, parents who are in harmonious marriages are more pleased with their relationship. Thus, they tend to view and participate in the developmental process of their children in a supportive manner. Such attitudes influence parenting behaviors and outcomes, ultimately acting on children's psychological adjustment, prosocial behavior, and other positive behavioral expression. From a conflict perspective, dissonance in mother-father relationships predicts children's cognitive control, psycho-emotional, and social development. In the cognitive control section, violence between parents can have some negative effects on executive functions, memorization, and other areas. In terms of psycho-emotional, children chronically in an atmosphere of marital conflict prefer to show anxiety and depression. In terms of social development, conflict behaviors between parents not only affect children's own adaptive and adjustment competence but also "spill over" into peer relationships. It is important to realize that marital conflict can not only affect social-emotional competence through direct effects but also negatively affect them in indirect ways through the home environment and educational process.

Previous studies can reveal the influence of marital conflict and support on social-emotional competence to some degree, but they lack dynamics and integration. This is demonstrated by the fact that marital conflict and support, being on two ends of mother-father relationships, both affect parents themselves and children to varying degrees. However, much of the research has been done from the perspective of relationship conflict, ignoring the value and positive impacts of marital support. At the same time, previous studies have mostly explored support and conflict as two separate constructs. They have rarely included both in the same perspective and considered their synergistic effects. Therefore, mother-father relationships could be considered from the dual perspectives of conflict and support.

1.4. The role of parental emotional expression and parent-child attachment in the link between mother-father relationships and social-emotional competence

1.4.1. Mother-father relationships-parental emotional expression-social-emotional competence

The study was interested in the role of the mediating variable of parental emotional expression first. As one of the pathways of parental emotion socialization, parental emotional expression is emotion-related expression styles or persistent patterns of communication that parents display in verbal or nonverbal form during family interactions. That is, mother-father relationships affect social-emotional competence by parental emotional expression. Harmonious and cordial mother-father relationships are usually accompanied by positive emotional expressions from the parents. This creates a warm and intimate family atmosphere, which finally affects the growth of prosocial behaviors and emotion regulation skills. Conversely, conflicts and breakdowns between parents usually trigger the venting of worse emotions in parents,

for example, anger, hostile emotions, etc. It may even lead to marital breakdown and eventually hinder the children's adjustment and lead to internalizing problem behaviors, etc (Schmidt et al., 2021).

According to the social learning mechanisms by which individual behavior occurs, individual behavior is not only influenced by the outcome of his or her actions but is also governed by a multitude of informative environmental cues. That is, parental emotional expression is influenced by the mother-father relationships. In other words, mother-father relationships create an environment for both parental emotional expression. This environment can more than induce, eliminate, and restore patterns of parental emotional expression. Mother-father relationships influence the relationship atmosphere between parents and even in the family. This positive or negative situation stimulates the emotional system of the father and mother, which in turn triggers different behavioral expressions, for example, positive or negative emotional expressions. That is, parents with discordant mother-father relationships exhibit more hostility (Chen et al., 2022). Thus, mother-father relationships are positively associated with parental emotional expression. Also, parental emotional expression is projected onto the next generation directly or indirectly, affecting the growth and development of children. Parents who have negative emotional expressions are willing to project stress and anxiety onto their children, causing them to experience more difficulties in the progression of emotional expression and social adjustment. Parental emotional expression could impact social-emotional competence on both positive and negative levels. On the positive section, children who live in a tolerant and understanding family atmosphere implicitly learn ways to express their emotions and to understand the emotions of others. This ultimately helps them to realize the improvement of social-emotional competence. From the negative section, prolonged exposure to conflicting, violent mother-father relationships can increase children's emotional insecurity. This in turn negatively affects an individual's competence to moderate their emotions and may even lead to psychological and behavioral issues such as depression and aggressive behaviors. Therefore, parental emotional expression is positively associated with social-emotional competence.

The family serves as a unit of emotional interaction in which parents interact as a process of emotional interaction (Huang et al., 2021). Not only do parents influence each other through emotional expression. In addition, they affect other family members through the socialization process, especially the emotional behavior of children, and their ability to develop. However, current research concentrates on the role and influence of parental emotional expression to some extent. Yet, previous studies have less considered issues related to the relationships between independent variables and dependent variables. Therefore, it is of great value to explore the mediating function of parental emotional expression in the link between mother-father relationships and social-emotional competence.

1.4.2. Mother-father relationships-parent-child attachment-social-emotional competence

We also considered parent-child attachment as a mediator. Parent-child attachment is a reciprocal, enduring emotional and physical bonding relationship between a child and the primary caregiver (Chung et al., 2020). It is considered a key mediating variable that has an important effect on the way that mother-father relationships influence social-emotional competence.

According to social learning theory, the degree to which individuals interact with the environment relates to the extent to which they are influenced by it. Namely, when children are actively interacting with mother-father relationships, they are influenced by the mother-father relationships which are seen as an environment. In turn, children can acquire their parents' language styles, information-processing strategies, and behavioral norms. Through this process, they can appropriately respond to their parents and express their love for them. Eventually, children will form a parent-child attachment with their

parents. That is, mother-father relationships can influence parent-child attachment. In other words, when mother-father relationships are high-quality, both fathers and mothers experience timely and appropriate emotional responses from the marriage, which in turn leads to feelings of pleasure, comfort, and happiness. And these positive affective experiences and cognitive patterns of interaction are transferable. Specifically, parents can transfer these to parent-child interactions, showing higher levels of acceptance and positive feedback to the child, thus building intimate and stable parent-child attachments. At the same time, studies have shown that the more conflict there is in the mother-father relationships, the more difficult it is to establish high-quality parent-child attachments. Therefore, there is a positive association between mother-father relationships and parent-child attachment.

In summary, current research mostly contributed to the effects and mechanisms of a single subsystem on other subsystems. Fewer studies are based on social learning theory and focus on the social learning among the mother-father subsystem, parent-child subsystem, and children's subsystem simultaneously. However, the development of social-emotional competence is a result of social learning. Therefore, we consider it important to capture the mediating function of parent-child attachment between independent and dependent variables based on a more comprehensive and systematic perspective.

1.4.3. Mother-father relationships-parental emotional expression-parent-child attachment-social-emotional competence

Although existing research has considered the function of parental emotional expression and parent-child attachment on social-emotional competence from different perspectives, the mechanisms by which they act jointly on social-emotional competence have not been made clear. Social learning theory may provide ideas for understanding the relationship between parental emotional expression and parent-child attachment. An individual's behavior is in a dynamic process of interaction with the situation. In this process, the relationship between parents serves as an external context that influences their emotional system. It stimulates the parental emotional expression first, which in turn affects their behavioral performance. And this behavioral expression is also reflected in the intimate interaction between parents and children. That is, parental emotional expression interacts with parent-child interaction behavior, which in turn affects parent-child attachment. In other words, parental emotional expression can positively predict parent-child attachment. The more positive emotions parents express in their interactions with children, the more sensitive they are to their children's emotional responses. This facilitates parents to treat their children in a supportive manner and promotes the formation of a virtuous attachment. Whereas, passive emotional expressions by parents in the family can lead to more frequent arguments or emotional behaviors that neglect the needs of children. Such behaviors can make children feel less emotionally secure. And they can cause children to reject intimate relationships with their parents. Therefore, the study concluded that the more positive the parental emotional expression, the higher the parent-child attachment. It should be noted that the function of parental emotional expression on parent-child attachment can be revealed not only through direct effects but also through indirect mediating effects (Batool and Lewis, 2020).

On one hand, parental emotional expression can mediate the effect of mother-father relationships on parent-child attachment. Parents in families with good mother-father relationships experience a lasting sense of well-being. They will show more positive emotions in the family, which will strengthen the parent-child attachment. Moreover, parents who are chronically in unhealthy relationships are more likely to spill over the unprocessed hostility and conflicts in mother-father relationships into the education of their children. It can result in harsh parenting, ignoring the children's inner feelings, and even threatening the children's sense of attachment security. On the other, parent-child attachment can also serve as a "bridge" to the effects of parental emotional expression on social-emotional competence. Specifically,

children will tend to obtain effective parent–child attachment with their parents who express positive emotions regularly in the home. Children are likewise more willing to accept and internalize their parents' rules and demands, imitate or acquire their parents' emotional expression patterns, and thus exhibit greater social-emotional competence (Lee et al., 2022). Conversely, parents can create a negative, stressful emotional atmosphere in the home that can hinder social-emotional competence.

However, both parental emotional expression and parent–child attachment can be considered different elements of “parenting”. “Parenting” is a potential mechanism that links independent and dependent variables (Lee et al., 2023). Nevertheless, existing studies on the mediating effects of parenting have used a narrow perspective such as parental emotion and parent–child relationship, which have certain limitations. Therefore, this research investigated several aspects of the parenting process as potentially mediating factors at the same time. Two variables, parental emotional expression, and parent–child attachment, were also included simultaneously in the effect of mother–father relationships on social-emotional competence. What's the function of mother–father relationships on social-emotional competence related to parental emotional expression and parent–child attachment? To what extent do parental emotion expression and parent–child attachment respectively mediate the influence of mother–father relationships on social-emotional competence? What are the co-mediating effects of both? These are still to be clarified. As a result, we attempted to explore the chain mediating influence of parental emotion expression and parent–child attachment in the link of social-emotional competence and mother–father relationships.

1.5. The present study

To sum up, current research has explored various elements' influence on social-emotional competence. However, fewer researches have included both parental emotional expression and parent–child attachment in the effect of mother–father relationships on social-emotional competence. Specifically, there are several major shortcomings in the existing relevant research. First, current research is less based on social learning theory and explores the synergistic effects of social learning in the family on children's social-emotional competence from the perspectives of imitative learning and alternative experiences. Second, most of the studies have been conducted from the perspective of marital conflict, and fewer studies have considered the mechanisms of the dual perspectives of marital support and marital conflict on social-emotional competence. Third, parental emotional expression in the family is an important pathway for children's social development. By contrast, less current research has been based on social learning points to consider the part of mother–father relationships in the improvement of social-emotional competence. Fourth, previous studies have mostly researched from a single subsystem, simultaneously focusing on the interaction of mother–father subsystem, parent–child subsystem, and children's subsystem. Fifth, there are huge cultural differences between Chinese and Western perceptions of parent–child attachment. However, fewer studies have been conducted to explore the potential mechanisms by which mother–father relationships affect social-emotional competence based on the Eastern background.

Therefore, based on social learning theory, the study aims to understand the potential mechanisms between mother–father relationships and social-emotional competence, using mother–father relationships as a starting point. In particular, we hypothesized that two mediating factors, parental emotional expression, and parent–child attachment, may jointly have chain mediating effects between mother–father relationships and social-emotional competence. That is, mother–father relationships can influence the improvement of social-emotional competence by the chain mediators of parental emotional expression and parent–child attachment. This study will verify this chain mediation model through a large-scale survey of Chinese children.

Specifically, we proposed the following three hypotheses in the present study (Fig. 1):

H1. Mother–father relationships would be significantly and positively correlated with social-emotional competence.

H2a. Parental emotional expression would mediate the relationship between mother–father relationships and social-emotional competence.

H2b. Parent–child attachment would mediate the relationship between mother–father relationships and social-emotional competence.

H3. Mother–father relationships would indirectly affect social-emotional competence through the chain mediating effect of parental emotional expression and parent–child attachment.

2. Methods

2.1. Participants

We selected kindergartens in Shaanxi, Guizhou, Shandong, Henan, and Jilin provinces of China by cluster sampling. Those provinces are in the northwest, southwest, eastern, and central regions of China. They all have a larger demographic foundation and a diversified population composition, which can represent all the populations in China. 802 children and their parents from 54 classes in 27 kindergartens were selected for this study. The sample consisted of 386 girls and 416 boys aged 37–87 months ($M = 2.25$, $SD = 0.77$), of whom 221 were 3–4 years old, 345 were 4–5 years old, and 236 were over 5 years old. Also, there were 621 one-child families and 181 two-child families. There were significant differences in the income and parents' education level of the sample families in different regions. However, the demographic information of the sample was not significantly different from that posted by the National Bureau of Statistics of China. Therefore, participants could be half of the total sample of age, gender, grade, and family status. Preschools provide three-year programs for children of 3 to 6 years old in China.

2.2. Measure

2.2.1. Mother–father relationships

To measure mother–father relationships, the Mother–Father Relationships Scale explored by Lee was used (Lee et al., 2023). It has been broadly used and has proven to be a reliable and valid measure (Baker et al., 2018). It consisted of two dimensions, marital support (5 items) and marital conflict (2 items), with a total of 7 items. The scale was accessed with a 5-point scale (1 = never to 5 = often). The results of the pre-survey in China showed that the Cronbach's alphas for this total scale and sub-scales were 0.76, 0.79, and 0.67 respectively, showing favorable internal consistency.

2.2.2. Parental emotional expression

To measure parental emotional expression, the Self-Expression in the Family Questionnaire was used. The scale was developed by Halberstadt and has been widely used in the assessment of parental emotional expression (Halberstadt et al., 1995). It is a parental report scale and consists of 40 items. These items include both positive emotional expression (23 items) and negative emotional expression (17 items) latitudes. It was accessed on a 5-point scale (1 = never, 5 = always). The results of the pre-survey in China showed that Cronbach's alphas for this total scale and sub-scales were 0.97, 0.92, and 0.95 respectively, showing favorable internal consistency.

2.2.3. Parent–child attachment

As to parent–child attachment, we used the Early Childhood Attachment Scale developed by Liu to measure it (Liu, 2005). It has been broadly used in China and has proven to be reliable (Zhu et al., 2018). The scale includes three dimensions of secure attachment (4 items),

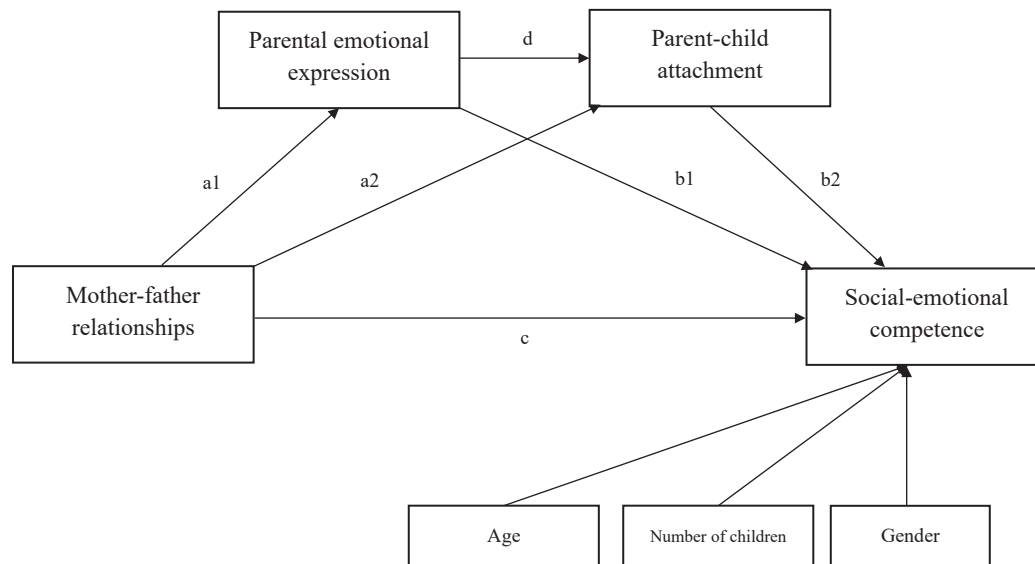


Fig. 1. A hypothesized chain mediation model.

ambivalent attachment (6 items), and avoidant attachment (11 items), with a total of 21 items. Parents accessed the items on a 5-point scale (1 = never, 5 = always). The Cronbach's alphas for the total scale and sub-scales were 0.88, 0.60, and 0.78 respectively.

2.2.4. Social-emotional competence

As for social-emotional competence, the study used the Chinese Inventory of Children's Social-Emotional Competence (CICSEC) explored by Li (Li et al., 2020a, 2020b). It has 4 sections. That is, emotion regulation (6 items), cognitive control (6 items), empathy and prosocial behaviors (11 items), and emotional expressivity (7 items). The scale has 30 items in total. It assesses the performance of children from 1 (strongly disagree) to 5 (strongly agree). The Cronbach's alphas for this total scale and sub-scales were 0.92, 0.88, 0.87, 0.90, and 0.83 respectively, showing favorable internal consistency.

2.3. Procedure

The materials and programs were agreed upon by Shaanxi Normal University. Kindergartens in several provinces of China, Shaanxi, Guizhou, Shandong, Henan, and Jilin, were chosen for us. The researchers obtained consent from the kindergarten administrators before entering the kindergartens. Upon entering the kindergartens, the researchers distributed informed consent forms to the participants. After obtaining consent, researchers invited mothers and fathers to fill in the scales of mother-father relationships, parental emotional expression, and parent-child attachment. Kindergarten teachers are paired to fill out the CICSEC based on the returned questionnaires. The survey lasted for 3 months. It is worth further explaining that this study mainly used offline filling out questionnaires to collect survey data. A total of 853 sets of questionnaires were distributed by the researchers. After excluding the questionnaires with incomplete answers and the research subjects with missing data on the main variables, the study obtained a total of 802 sets of valid questionnaires. The effective recovery rate and completion rate of the questionnaires were 94.02%.

2.4. Statistical analyses

First, we used the SPSS 25 application to explore descriptive statistics and correlations. Second, we utilized the AMOS 24 application to develop the Structural Equation Model to research the mediating effect of parental emotional expression and parent-child attachment in the

link between mother-father relationships and social-emotional competence. We tested the fit of the model and accessed hypotheses based on the causality model. Finally, the study utilized SPSS 25 to deal with the collected data. When constructing the Structural Equation Model, the study included control variables (gender, age, number of children) in the model to obtain a more precise test of the link between the variables.

3. Results

3.1. Descriptive data and correlations

In Table 1, descriptive statistics and correlations between the sections of mother-father relationships, parental emotional expression, parent-child attachment, and social-emotional competence were shown. As we know, social-emotional competence was positively associated with marital support, positive emotional expression, and secure attachment. At the same time, social-emotional competence was negatively correlated with marital conflict, negative emotional expression, ambivalent attachment, and avoidant attachment. Furthermore, age was considered to be correlated with marital conflict positively and significantly. This indicates that the older the children were, the more frequent the parental marital conflict was. Furthermore, positive emotional expression and social-emotional competence were positively correlated with gender, respectively. This indicates that positive emotional expression and social-emotional competence were significantly different by child gender.

3.2. Mediation analyses

Existing studies have found that gender, age, and the number of children were significantly different in social-emotional competence. As a result, the study controlled for the above background variables. Among these, the study set gender and the number of children as dummy variables (male = 0, female = 1; one child in the home = 0, two children in the home = 1). The outcomes of model fitting presented that in the predictive model of mother-father relationships on social-emotional competence, only gender showed a significant predictive coefficient on social-emotional competence. In other words, the number of children, age, and social-emotional competence weren't significantly correlated. And gender was significantly associated with social-emotional competence. That is, girls have greater social-emotional competence compared to boys. From the point of physiological

Table 1
Descriptive statistics and correlations among variables.

	1	2	3	4	5	6	7	8	9	10	11
1.Age	–										
2.Number of children	0.09***	–									
3.Gender	–0.05	0.03	–								
4.Marital support	0.01	–0.02	0.05	–							
5.Marital conflict	0.07***	–0.03	–0.02	–0.35***	–						
6.Positive emotional expression	–0.01	–0.03	0.07***	0.50***	–0.37***	–					
7.Negative emotional expression	0.01	0.01	–0.05	–0.47***	0.36***	–0.81***	–				
8.Secure attachment	0.02	–0.02	–0.01	0.46***	–0.42***	0.66***	–0.62***	–			
9.Ambivalent attachment	0.01	–0.04	–0.03	–0.48***	0.41***	–0.66***	0.77***	–0.68***	–		
10.Avoidant attachment	0.03	–0.02	–0.02	–0.47***	0.37***	–0.77***	0.74***	–0.67***	0.73***	–	
11.Social-emotional competence	0.01	–0.01	0.08***	0.55***	–0.44***	0.88	–0.84***	0.76***	–0.79***	–0.84***	–
M	2.25	1.75	1.48	4.03	4.07	3.70	3.67	3.86	3.81	3.62	3.25
SD	0.77	0.50	0.50	0.71	0.51	0.58	0.53	0.53	0.51	0.48	0.72

Note. N = 802. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

developmental characteristics, girls are more sensitive to interpersonal interactions. It means that girls are better able to capture valid messages from parental interactions and are more susceptible to the influence of parents. These will contribute to the improvement of girls' social-emotional competence. Therefore, the gender variable was controlled for in this study.

3.2.1. A test of the direct effects of mother-father relationships on social-emotional competence

In Fig. 2, there was a model test of the direct effects of mother-father relationships on social-emotional competence. This findings showed a great model fit ($\chi^2 = 13.194$, $df = 8$, $\chi^2/df = 1.65$, RMSEA = 0.028, GFI = 0.995, AFGI = 0.986, RFI = 0.993, NFI = 0.997, PNFI = 0.531). In Table 2, we could find support from the data that mother-father relationships ($\beta = 0.33$, $SE = 0.08$, $t = 8.91$, $p < 0.001$) could predict social-emotional competence significantly and positively, which supported Hypothesis 1.

3.2.2. A test of the indirect effects of mother-father relationships on social-emotional competence

The model included four latent variables: mother-father relationships, parental emotional expression, parent-child attachment, and social-emotional competence. The four variables were set correlated in this measurement model the factor loadings were freely estimated, and the measurement model was found to fit well ($\chi^2 = 67.03$, $df = 40$, $\chi^2/df = 1.68$, RMSEA = 0.036, RFI = 0.947, GFI = 0.932, IFI = 0.966, NFI = 0.962, PNFI = 0.699), allowing for subsequent full model analysis. In Fig. 3, the full model tests for the four latent variables are shown. The findings found great full model fit indices ($\chi^2 = 98.39$, $df = 38$, $\chi^2/df = 2.59$, RMSEA = 0.042, RFI = 0.951, CFI = 0.970, NNFI = 0.957, PNFI = 0.667, PGFI = 0.541).

A deeper analysis of the model paths showed significant path coefficients for both parental emotional expression ($\beta = 0.50$, $SE = 0.02$, $t = 7.00$, $p < 0.001$) and parent-child attachment ($\beta = 0.47$, $SE = 0.05$, $t = 4.87$, $p < 0.001$) on social-emotional competence. This path coefficients of mother-father relationships on parental emotional expression ($\beta = 0.81$, $SE = 0.16$, $t = 8.92$, $p < 0.001$) and parent-child attachment ($\beta = 0.31$, $SE = 0.05$, $t = 3.43$, $p < 0.001$) were both positive, but the path coefficient on dependent variable got negative. Based on this, we could realize that the predictive effect of mother-father relationships on social-emotional competence is fully mediated by parental emotional expression and parent-child attachment. Also, the path coefficients of parental emotional expression ($\beta = 0.69$, $SE = 0.01$, $t = 7.01$, $p < 0.001$) on parent-child attachment were significant, indicating the presence of a chain mediator of parental emotional expression and parent-child attachment.

Table 2
Standardized estimates for direct and indirect effects of variables.

Path	β	SE	95%CI	
			Lower	Upper
Direct effects				
MFR $\rightarrow$ PEE(a1)	0.81***	0.16	0.46	0.66
MFR $\rightarrow$ PCA(a2)	0.31***	0.05	0.27	0.47
PEE $\rightarrow$ PCA(d)	0.69***	0.01	0.31	0.35
MFR $\rightarrow$ SEC(c)	0.33***	0.08	0.26	0.57
PEE $\rightarrow$ SEC(b1)	0.50***	0.02	0.47	0.56
PCA $\rightarrow$ SEC(b2)	0.47***	0.05	0.46	0.70
Indirect effects				
MFR $\rightarrow$ PEE $\rightarrow$ SEC(a1*b1)	0.41***	0.02	0.24	0.32
MFR $\rightarrow$ PCA $\rightarrow$ SEC(a2*b2)	0.15***	0.01	0.05	0.09
MFR $\rightarrow$ PEE $\rightarrow$ PCA $\rightarrow$ SEC(a1*d*b2)	0.26***	0.02	0.14	0.21

Note. N = 802. MFR = Mother-father relationships, PEE = Parental emotional expression, PCA = Parent-child attachment, SEC = Social-emotional competence, β = standardized estimates, CI = confidence interval, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

By utilizing the Bootstrap way of Bias-Corrected Bootstrap with randomized repetitive samples of 5000 for estimation, the significance

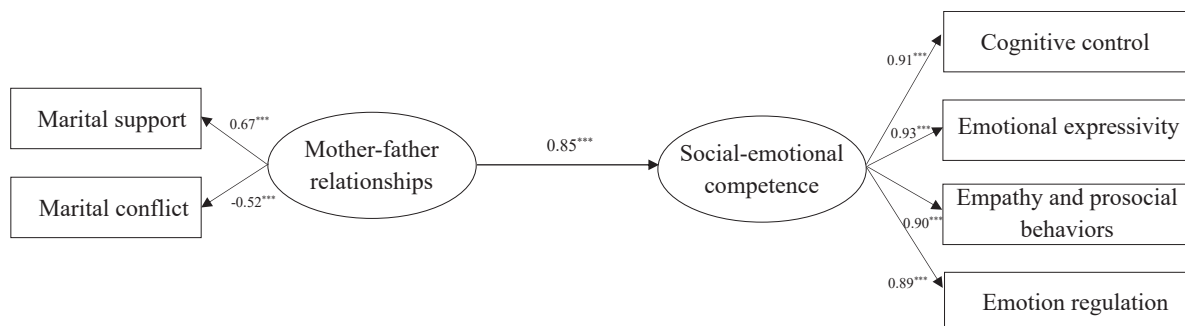


Fig. 2. The standardized path coefficients describe the direct effects of mother-father relationships on social-emotional competence. *** $p < 0.001$.

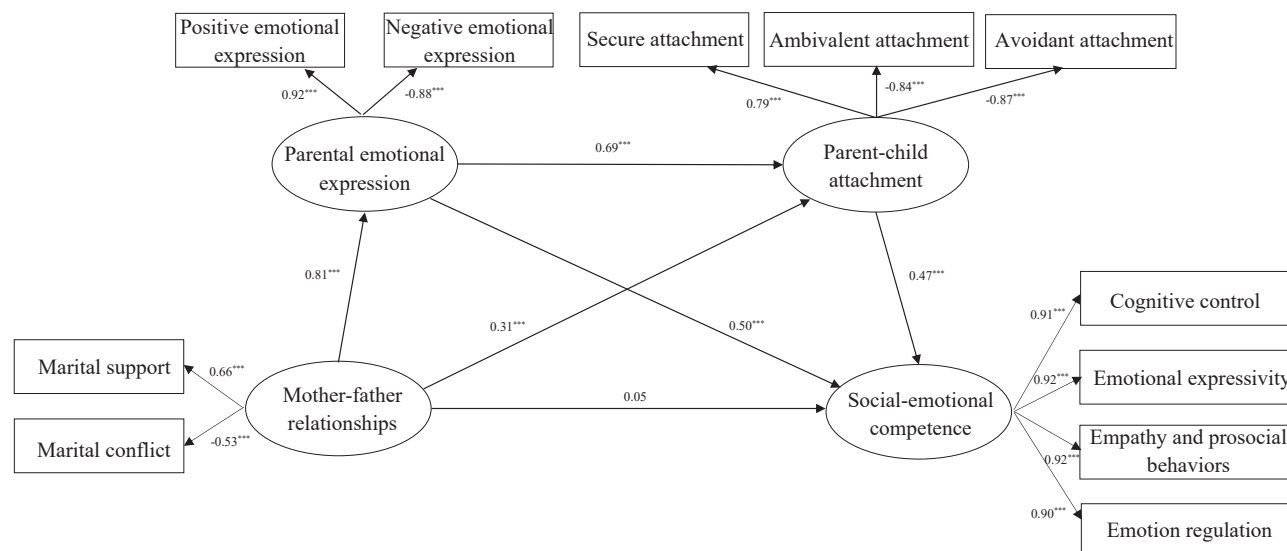


Fig. 3. The standardized path coefficients describe the indirect effects of mother-father relationships on social-emotional competence. *** $p < 0.001$.

of the structural model mediation influence was tested. The 95% confidence interval for each indirect influence test didn't have 0 and was significant statistically in Table 2. Therefore, the chain mediating influence of parental emotional expression and parent-child attachment on the dependent variable was positive, which supported Hypothesis 2a/2b/3.

4. Discussion

To sum up, this study had 3 main findings. Firstly, mother-father relationships were positively associated with social-emotional competence, which verified Hypothesis 1. Secondly, parental emotional expression was a mediator between the link of the independent variable and dependent variable, which verified Hypothesis 2a. The parent-child attachment was a mediator of the association between the independent and the dependent variables, which verified Hypothesis 2b. Finally, parental emotional expression and parent-child attachment acted as chain mediators between the independent and the dependent variables, which verified Hypothesis 3. Overall, the results supported the primary hypotheses of the study. Furthermore, the study revealed that parental emotional expression and parent-child attachment were at the center of the model. Parents can promote social-emotional competence through active emotional expression and high-quality parent-child attachment in addition to operating good mother-father relationships. The result is in agreement with previous studies in other domains that interactive relationships and interactive behaviors in families can interact to jointly effect growth (Martí et al., 2016).

4.1. The direct effects of mother-father relationships on social-emotional competence

We found that mother-father relationships significantly predicted social-emotional competence, which verified Hypothesis 1. That is, the more trusting the mother-father relationships, the greater the social-emotional competence.

It is in keeping with previous studies. This means that mother-father relationships, consisting of marital support and conflict, can influence social-emotional regulation. That is, the more marital support there is in the mother-father relationships, the more relaxed and secure children will feel in the family. They are more willing to share and express their thoughts and feelings, and their social-emotional competence develops better. Conversely, the more marital conflict there is, the more vulnerable children are to harm from their fathers and mothers, and they

become fearful, sensitive, and helpless. All of these can negatively affect children's cognitive control, and empathy (Espejo-Siles et al., 2020). Children and parents interplay with mutual interaction and the relationship between them is bidirectional (Elliott and Davis, 2020). That is the more harmonious the mother-father relationships, the more the parents concentrate on the children. Children are more likely to give back to their families. They transfer these positive feelings and behaviors to their surroundings and even to the larger social environment and develop better social-emotional competence.

Also, this study supports the individual-environment matching review of social learning theory within the perspective of mother-father relationships. That is, children's behavior and psychology are shaped by both the individual and the context. It is affected by both the external environment and its internal factors and the relationship between them (Su et al., 2015). And mother-father relationships, as external environmental stimuli experienced by children, are an important influence on the improvement of social-emotional competence. When marital conflict is frequent, the child in a negative external environment usually has difficulty acquiring good emotion regulation strategies, interaction and communication skills, etc. This could impede the progression of social-emotional competence. This suggests that mother-father relationships are an important antecedent variable influencing social-emotional competence. It offers novel points for improving social-emotional competence. That is, in addition to focusing on the children's factors, a harmonious and positive emotional climate can also be created to assist children in acquiring social-emotional competence through external factors such as cordial mother-father relationships. In addition, the child with certain susceptibility features is not only susceptible to maladjustment due to the adverse effects of an unfavorable environment. Children are also susceptible to favorable influences of positive environments that lead to better development. That is, the more children are in an atmosphere of positive mother-father relationships and their emotional expression, the better the acquisition of social-emotional competence, and vice versa.

Significantly, previous research has found that there may be more complex mechanisms between independent and dependent variables. In particular, the influence may become insignificant when other behavioral or relational interactions in the family are added. This suggests that the link is valid at all and may require other factors to work.

4.2. The mediating effect of parental emotional expression and parent–child attachment

4.2.1. The mediating effect of parental emotional expression

These findings suggested that parental emotional expression was a mediator between the link of independent variables and dependent variables. Hypothesis 2a was tested. In other words, parental emotional expression can reflect the emotional experiences between parents. It is an important mediating variable in the “spillover” of mother–father relationships to social-emotional competence.

This further supports the previous studies. Specifically, a cordial, harmonious relationship between parents can be expected to result in more positive expressions of emotion, such as expressing gratitude and providing comfort (Meyer et al., 2014). Children can realize how to project emotions subtly by imitating and learning appropriate emotion expressions and effective emotion regulation strategies. All of these can support to achievement of early development of social-emotional competence. Conversely, the more marital conflict a couple perceives, the more negative emotions they express in their family life. It may lead a child to perceive destructive arguments and disagreements between parents and produce more social avoidance behaviors. Children also acquire inappropriate emotional paradigms through the “mirroring” process, which ultimately hinders their social-emotional development (Qiu and Shum, 2022).

We can use the emotional security hypothesis to explain this phenomenon (Schudlich et al., 2019). The hypothesis suggests that high-quality mother–father relationships promote active emotions and the expression of parents. Children gain a sense of security when they perceive a close and stable parent–child relationship. This positively affects the social-emotional competence. That is, greater or worse mother–father relationships directly affect parental emotional socialization behaviors in the family. In turn, emotional expression is both one of the forms of parental emotional socialization expression and a reference paradigm for children to engage in emotional expression. Good parental emotional expression enhances children’s emotional security and leads to a positive pattern of interpersonal interactions. This finally improves social-emotional competence.

Similarly, according to the substitution hypothesis, mother–father communication can to some extent substitute for parent–child emotional interaction (Coyne et al., 2014). In cases where there is more marital conflict between parents, both parents tend to be immersed in the conflict between the couple. At this time, parents will neglect emotional expression and interaction in the home and express less emotion in the home environment. This can leave children in a stressful, depressing family environment that reduces their reverence and trust in their parents, and also exposes them to the great danger that their relationship may break down at any time. It will result in difficulties for children to develop social-emotional competence.

4.2.2. The mediating effect of parent–child attachment

We found that parent–child attachment was also a mediator connecting the changes between mother–father relationships and social-emotional competence, which verified Hypothesis 2b. That is to say, with the inclusion of mediating variables, mother–father relationships do not affect social-emotional competence directly. Instead, there is an indirect effect between the independent variable and dependent variable through different attachment relationships.

This result complements previous studies. Specifically, families with more marital support tend to have a more harmonious relationship between their parents. They can provide children with good parenting experiences that lead to quality parent–child attachments. Children will have more confidence to feel the flow of information between different individuals and develop prosocial behaviors such as caring and mutual support, humility, and sharing. It influences the acquirement of social-emotional competence. Conversely, in families with more marital conflict, parents tend to keep each other at arm’s length. They are more

distant, cold, and conflicted with each other. And this distance can be reflected in parent–child interactions. This tends to keep children in conflict and tension, weakening the emotional bond between parent and child and making it difficult to form high-quality parent–child attachment. Children also do not receive positive feedback from parent–child interactions and learn the correct way to interact with others, which ultimately results in the underdevelopment of social-emotional skills. Moreover, interaction and attachment behaviors in parent–child relationships can serve as “learning scaffolds” for children’s interactions in social relationships outside the family. When quality parent–child attachment is lacking, parents are unable to construct a learning environment for children’s social interactions through effective parent–child interactions, especially the communication of internal psychological states. This affects the acquisition of desirable social habits and behaviors and makes it difficult to provide “scaffolding” for children’s comprehension of their own and others’ social activities. These can ultimately lead to problems in the improvement of social-emotional competence.

Meanwhile, the findings could be accounted for using the inner workings model theory of attachment. Parents in families with more marital support concentrate on children’s needs, and the child will be willing to form secure attachments with his or her parents. It will develop positive internal working patterns for processing external stimuli in long-term interactions. This model allows children to have good emotional processing and to understand the psychology and behaviors of others. In turn, children are better able to engage in dialogue with parents or others and achieve early development of social-emotional competence. Conversely, for children, it is difficult to cultivate great internal working patterns and reap the developmental benefits of quality parent–child relationships (Zhao et al., 2020). It will ultimately hinder the improvement of social-emotional competence.

4.3. The chain mediation effect of parental emotional expression and parent–child attachment

This research found that parental emotional expression and parent–child attachment both were chain mediators between the independent variable and dependent variable. Hypothesis 3 was verified. That is, the effect of mother–father relationships on social-emotional competence will not be remarkable after the introduction of two mediating variables, parental emotional expression and parent–child attachment. This suggests that mother–father relationships are a distal variable. Its impact on social-emotional competence could be better achieved through the “bridging” function of parental emotional expression and parent–child attachment.

In terms of family interactions, interpersonal interactions among family members are important in shaping individual variations in social-emotional competence. Family interactions mainly include both relational and behavioral aspects. The former means that mother–father relationships and the parent–child relationships formed by fathers and mothers, child and parents in family interactions, etc. The latter are emotions and attitudes that are expressed through family interactions. Nevertheless, existing research has not worked out the mechanisms by which relational and behavioral interactions in families collectively influence the improvement of social-emotional competence. As a result, the finding complements previous findings. Specifically, parents with more marital support were willing to show active emotions in their families. This will empower parents to engage in parent–child interactions sensitively and cooperatively. Mothers and fathers are also more likely to have a high-quality attachment with their children. In the process, parents will guide children to develop friendly, cooperative, and unprincipled social patterns continuously to enhance their social-emotional competence. However, mother–father relationships with more marital conflict may result in parents depleting their psychological and emotional resources. Parents are unconsciously irritable and impatient in dealing with family conflicts, exposing more negative

emotions. This can force parents to bring the tensions of the mother–father relationships into the parent–child interaction. Consequently, parents will be unable to have an emotionally secure attachment to their children. This can further affect children’s perception of their own emotions and increase the risk in their social-emotional development.

It can be interpreted by social learning theory. Children imitate their parents’ interaction behaviors in marriage and carry the acquired behaviors into parent–child interactions or other interactions outside the home. This can further affect parent–child relationships or social interactions (Li et al., 2020a, 2020b). That is to say, children are permanently in the home atmosphere with more marital support, where parents can express their emotions appropriately. Therefore, parent–child interactions are frequent and attachment is high-quality. Children can follow the model of their parents and receive adequate parenting support from them. As time passes, children also gradually acquire this pattern of family interaction and the necessary social skills. And, children transfer this to social interactions with others, promoting the improvement of their social-emotional competence. This explanation based on the triangular link of family structure is also relevant. If parents in the family interact more frequently and express more emotions, children will become involved. Triangles are formed in the family, either actively or passively, to relieve parental stress and to keep the family system in balance. In such relationships, parents with more marital conflict shift their focus from the arguments and disagreements in the binary relationship to the negative interactions between parents and children. In turn, negative parental emotional expression can damage the parent–child relationship. This can place children in a high-pressure, anxious family emotional climate, further hindering the formation of high-quality attachment relationships. During this process, children have difficulty acquiring effective emotion management strategies, exhibit less helpful behaviors, and continually weaken their sense of well-being and attachment. This will negatively affect the acquirement of social-emotional competence.

To sum up, the present study was based on social learning theory and was conducted in terms of the interaction dynamics between children and parents. The research wants to explore more deeply the influence of relational and behavioral interactions at home on social-emotional competence. It is relatively groundbreaking. Moreover, the association between relational interactions, behavioral interactions, and social-emotional competence will be different as the child develops. Based on these, the study presumes that a feedback loop influence may exist between relational interactions, behavioral interactions, and children’s development. That is to say, it has a “reciprocal” link between relational interactions, behavioral interactions, and children’s development. For example, relational and behavioral interactions contribute to the improvement of social-emotional competence. Correspondingly, they are also affected by the developmental changes in children to offer better family relational and behavioral interactions for children.

5. Conclusions, limitations, and implications

The study provides more worthwhile insertions into how the independent variables influence dependent variables. Firstly, mother–father relationships significantly and positively predicted social-emotional competence. Secondly, both parental emotional expression and parent–child attachment are mediators in the link from the independent variable to the dependent variable. Thirdly, parental emotional expression and parent–child attachment played the parts of chain mediating in the link between the independent variable and dependent variable.

Yet, the research also has some limitations. Firstly, the impact of mother–father relationships on social-emotional competence is a dynamic process. As a cross-sectional study, this study can’t be in a position that examines the characteristics and causal relationships of the variables over time. Therefore, future studies may consider using a longitudinal research design to more effectively explain the substance of

the link between the independent variable and the dependent variable. Secondly, this study used parents’ self-reports. The reported results may affect the ideal situation that participants tend to express. Among these, undesirable behaviors are frequently underestimated, while desirable behaviors are exaggerated. Therefore, future research can combine other assessment models and use multiple measures and sources of information to corroborate each other and thus construct a more comprehensive evaluation measure. Thirdly, the study did not distinguish between fathers’ and mothers’ response results in the collection of self-reports. However, the interdependence of parents is important. Therefore, future studies could also collect paternal and maternal data separately. This integration of the perspectives of both mothers and fathers in the home in the same framework could result in a better realization of the mechanisms by which mother–father relationships impact social-emotional competence. Finally, children are the most active part of the family. There is likely an interaction between social-emotional competence and mother–father relationships, parental emotional expression, and parent–child attachment. Therefore, it is important to explore the contribution of social-emotional competence in promoting positive parenting and improving family relationships in future research.

Despite some limitations, we also have some potential innovations. In terms of the view of theoretical, this study broadens the usage of the social learning theory of the study related to social-emotional competence. Meanwhile, we find an intrinsic link between different elements of parenting and social-emotional competence, providing theoretical guidance for social-emotional competence improvement. From a practical standpoint, this study provides new ideas to develop social-emotional competence. Firstly, mothers and fathers should concentrate on the cultivation of social-emotional competence and provide diverse learning opportunities for children. Secondly, parents should be deeply aware of the importance of mother–father relationships to social-emotional competence. Thus, they should strive to manage mother–father relationships, enhance the couple’s emotions, and reduce family conflicts. These further create a warm family atmosphere, maintain a harmonious marital relationship, and develop social-emotional competence. Thirdly, reasonable emotional expression is one of the mandatory courses for parents. Meanwhile, mothers and fathers are expected to contribute to children’s psychological changes in their interactions with them and provide them with high-quality support and companionship. In this process, parents can allow children to experience care and warmth in the family, providing a good environment for improving social-emotional competence.

6. Ethics statements

All surveys of the study complied with the 2013 Declaration of Helsinki and its later editions or equivalent standards of ethics.

Funding

This study was supported by “The Institutional Design and Longitudinal Study of Family Education Compensation for De Facto Unattended Children (BKA220039)”.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Baker, E. A., DeLuca Bishop, H., Stigall, L. A., & van Dulmen, M. H. M. (2018). Positive parental engagement: Investigating the role of the mother-father relationship. *Journal of Family Psychology*, 32(8), 1005–1014.
- Bandura, A., & Walters, R. H. (1977). *Social learning theory* (Vol. 1). Englewood Cliffs: Prentice Hall.
- Batool, S. S., & Lewis, C. A. (2020). Does positive parenting predict pro-social behavior and friendship quality among adolescents? Emotional intelligence as a mediator. *Current Psychology*, 1–15.
- Brady, W. J., McLoughlin, K., Doan, T. N., & Crockett, M. J. (2021). How social learning amplifies moral outrage expression in online social networks. *Science. Advances*, 7(33), eabe5641.
- Chen, B. B., Qu, Y., Yang, B., & Chen, X. (2022). Chinese mothers' parental burnout and adolescents' internalizing and externalizing problems: The mediating role of maternal hostility. *Developmental psychology*, 58(4), 768.
- Chung, G., Lanier, P., & Wong, P. Y. J. (2020). Mediating effects of parental stress on harsh parenting and parent-child relationship during coronavirus (COVID-19) pandemic in Singapore. *Journal of family violence*, 1–12.
- Coyne, S. M., Padilla-Walker, L. M., Fraser, A. M., Fellows, K., & Day, R. D. (2014). "Media time= family time" positive media use in families with adolescents. *Journal of Adolescent Research*, 29(5), 663–688.
- Cundiff, J. M., Smith, T. W., Butner, J., Critchfield, K. L., & Nealey-Moore, J. (2015). Affiliation and control in marital interaction: Interpersonal complementarity is present but is not associated with affect or relationship quality. *Personality and Social Psychology Bulletin*, 41(1), 35–51.
- Elliott, S., & Davis, J. M. (2020). Challenging taken-for-granted ideas in early childhood education: A critique of Bronfenbrenner's Ecological Systems Theory in the age of post-humanism. *Research handbook on childhood nature: Assemblages of childhood and nature research*, 1119–1154.
- Espejo-Siles, R., Zych, I., Farrington, D. P., & Llorent, V. J. (2020). Moral disengagement, victimization, empathy, social and emotional competencies as predictors of violence in children and adolescents. *Children and Youth Services Review*, 118, Article 105337.
- Halberstadt, A. G., Cassidy, J., Stifter, C. A., Parke, R. D., & Fox, N. A. (1995). Self-expressiveness within the family context: Psychometric support for a new measure. *Psychological assessment*, 7(1), 93–103.
- Huang, Q., Sun, J., & Tang, Y. (2021). Chinese parents' scaffolding and children's initiative in mother-child and father-child interactions across different types of problem-solving activities. *Early Education and Development*, 32(2), 249–271.
- Lavner, J. A., Karney, B. R., & Bradbury, T. N. (2016). Does couples' communication predict marital satisfaction, or does marital satisfaction predict communication? *Journal of Marriage and Family*, 78(3), 680–694.
- Lee, J. K., Marshall, A. D., & Feinberg, M. E. (2022). Parent-to-child aggression, intimate partner aggression, conflict resolution, and children's social-emotional competence in early childhood. *Family Process*, 61(2), 823–840.
- Lee, Y., Kim, K., Zeng, S., & Douglass, A. (2023). Mother-father relationships and child social-emotional adjustment: Mediation through maternal and paternal parenting. *Early Childhood Research Quarterly*, 63, 15–23.
- Li, C., Jiang, S., Fan, X., & Zhang, Q. (2020a). Exploring the impact of marital relationship on the mental health of children: Does parent-child relationship matter? *Journal of Health Psychology*, 25(10–11), 1669–1680.
- Li, X., Lam, C. B., Chung, K. K. H., Cheung, R. Y. M., Leung, C., & Fung, W. K. (2020b). Development and validation of the Chinese Inventory of Children's Socioemotional Competence (CICSEC). *Early Education and Development*, 31(6), 854–872.
- Li, S., Tang, Y., & Zheng, Y. (2023). How the home learning environment contributes to children's social-emotional competence: A moderated mediation model. *Frontiers in Psychology*, 14, 491.
- Liu, J. M. (2005). *Attachment of young children, The relationship between language and interpersonal intelligence and mental*. Graduate School of Education, National Chengchi University. Master's Thesis.
- Martí, M., Bonillo, A., Jané, M. C., Fisher, E. M., & Duch, H. (2016). Cumulative risk, the mother-child relationship, and social-emotional competence in Latino Head Start children. *Early Education and Development*, 27(5), 590–622.
- Meyer, S., Raikes, H. A., Virmani, E. A., Waters, S., & Thompson, R. A. (2014). Parent emotion representations and the socialization of emotion regulation in the family. *International Journal of Behavioral Development*, 38(2), 164–173.
- Qiu, C., & Shum, K. K. M. (2022). Relations between caregivers' emotion regulation strategies, parenting styles, and preschoolers' emotional competence in Chinese parenting and grandparenting. *Early Childhood Research Quarterly*, 59, 121–133.
- Salali, G. D., Chaudhary, N., Bouer, J., Thompson, J., Vinicius, L., & Migliano, A. B. (2019). Development of social learning and play in BaYaka hunter-gatherers of Congo. *Scientific Reports*, 9(1), 11080.
- Schmidt, A., Kramer, A. C., Brose, A., Schmiedek, F., & Neubauer, A. B. (2021). Distance learning, parent-child interactions, and affective well-being of parents and children during the COVID-19 pandemic: A daily diary study. *Developmental Psychology*, 57(10), 1719.
- Schudlich, T. D. D. R., Jessica, N. W., Erwin, S. E., & Rishor, A. (2019). Infants' emotional security: The confluence of parental depression, Interparental conflict, and parenting. *Journal of Applied Developmental Psychology*, 63, 42–53.
- Su, R., MEvang, A., & Øverlien, C. (2015). 'If you look, you have to leave': Young children regulating research interviews about experiences of domestic violence. *Journal of Early Childhood Research*, 13(2), 113–125.
- Zhao, F., Liu, M., & Li, S. (2020). Paternal coparenting behavior and adolescent prosocial behaviors: Roles of parent-child attachment, peer attachment, and gender. *Children and Youth Services Review*, 119, Article 105629.
- Zhu, J. J., Zhang, Y., Li, Y., Peng, R. Z., & Feng, M. (2018). Maternal Attachment Style and Child Anxiety: The Mediating Role of Parent-child Relationship. *Chinese Journal of Clinical Psychology*, 04, 763–766.